

Connective	Symbol	Example (p and q are simple statements)
p and q	$\wedge$	$p \wedge q$
p or q	$\vee$	$p \vee q$
not p	$\neg$ or $\sim$	$\neg p$ or $\sim p$
Implies (if then)	$\rightarrow$	$p \rightarrow q$
if and only if (iff)	$\leftrightarrow$	$p \leftrightarrow q$

The **truth value** of a proposition is true, denoted by **T**, if it is a **true proposition**, and the truth value of a proposition is **false**, denoted by **F**, if it is a **false proposition**.

It uses truth table to get **semantics of the logical formulae**.

A	B	$\neg A$	$A \wedge B$	$A \vee B$	$A \rightarrow B$	$A \leftrightarrow B$
1	1	0	1	1	1	1
1	0	0	0	1	0	0
0	1	1	0	1	1	0
0	0	1	0	0	1	1

- $A \rightarrow B$ is known as **modus ponens**.
- By inspecting the table, we can see that $A \rightarrow B$ is equivalent to $\neg A \vee B$ and that $A \leftrightarrow B$ is equivalent to $(A \rightarrow B) \wedge (B \rightarrow A)$. (The latter is also equivalent to $\neg(A \oplus B)$, where $\oplus$ is exclusive-or.)

Conjunctive Normal Form (CNF)

Also known as Product of Sums (POS). A Boolean formula is in CNF if it is a conjunction (AND) of one or more clauses, where each clause is a disjunction (OR) of literals.

- **Literal:** A Boolean variable (P) or its negation ($\neg P$).
- **Clause:** A set of literals connected purely by logical OR ($\vee$).
- **Structure:** AND of ORs.

Example:

$$(A \vee B) \wedge (\neg C \vee D \vee E) \wedge (\neg A)$$

Disjunctive Normal Form (DNF)

Also known as Sum of Products (SOP). A Boolean formula is in DNF if it is a disjunction (OR) of one or more clauses, where each clause is a conjunction (AND) of literals.

- **Literal:** A Boolean variable (P) or its negation ($\neg P$).
- **Clause:** A set of literals connected purely by logical AND ($\wedge$).
- **Structure:** OR of ANDs.

Example:

$$(A \wedge B) \vee (\neg C \wedge D \wedge E) \vee (\neg A)$$

Key Properties

- **Equivalence:** Every propositional logic formula can be converted into an equivalent CNF and DNF representation using De Morgan's Laws, double negation elimination, and distributive laws.
- **Applications:**
 - **CNF:** The standard input format for Boolean Satisfiability (SAT) solvers and automated theorem proving.
 - **DNF:** The standard format for deriving logic circuits directly from truth tables in digital electronics.

De Morgan's Laws are a pair of fundamental transformation rules in Boolean algebra and set theory. They describe how mathematical statements and their opposites relate to one another when dealing with logical AND/OR operations or set unions/intersections.

In simple terms, they state that applying a negation (NOT) to a grouped logical operation flips the operators (AND becomes OR, OR becomes AND) and distributes the negation to each individual term.

1. In Boolean Logic

Let A and B be Boolean variables.

- **Law 1: The negation of a conjunction is the disjunction of the negations.**

$$\neg(A \wedge B) \iff (\neg A) \vee (\neg B)$$

(NOT (A AND B) is equivalent to (NOT A) OR (NOT B))

- **Law 2: The negation of a disjunction is the conjunction of the negations.**

$$\neg(A \vee B) \iff (\neg A) \wedge (\neg B)$$

(NOT (A OR B) is equivalent to (NOT A) AND (NOT B))

Premises: $P \rightarrow Q$, $Q \rightarrow R$, P . Conclusion: R (Check validity)

Show thinking ▼

Validity: Valid

Proof using Rules of Inference:

1. $P \rightarrow Q$ (Premise)
2. $Q \rightarrow R$ (Premise)
3. P (Premise)
4. Q (Modus Ponens using 1 and 3)
5. R (Modus Ponens using 2 and 4)

Since the conclusion R is logically derived from the given premises, the argument is valid.

Alternatively, using **Hypothetical Syllogism**:

1. $P \rightarrow Q$ and $Q \rightarrow R$ logically implies $P \rightarrow R$.
2. Given $P \rightarrow R$ and the premise P , R must be true (Modus Ponens).